FINANCIAL RISK PLATFORM

Five Synchronized Frontend / Backend Workstreams

The project is organized into five paired workstreams. Each pair has a frontend implementation plan and a backend service/API plan. This index defines the common architecture and delivery sequence.

The intended MVP is broad in platform coverage but lightweight in initial data volumes, integrations, UI/UX depth and domain-model completeness. It should demonstrate the architecture and a few representative end-to-end capabilities without pretending every financial module is production validated.

Shared four-layer architecture

Layer	Responsibility
1. Universal Platform Foundation	Data ingestion, canonical schema and metadata, data quality, identity/access, audit, versioning and change governance.
2. Configurable Financial Domain	Instrument definitions, mappings, rule registry, assumptions, configuration lifecycle and calculation-module contracts.
3. Financial Calculation Modules	Liquidity/LCR/HQLA, cash-flow modelling, maturity ladder, and extension points for ALM/IRRBB, collateral and capital.
4. Intelligence and Experience	Natural-language proposals, assistant, scenarios, lineage, explainability, agentic workflows and evaluation.

Five paired workstreams

Pair	Scope	Why it exists
WS1	Data Governance, Security & Platform Foundation	Establish a trustworthy, secure, versioned foundation across all financial modules.
WS2	Data Ingestion, Mapping & Data Quality	Accept uncertain source shapes safely through configurable contracts, mappings and validation rather than hardcoding one client file format.
WS3	Financial Domain Model & Calculation Services	Provide reusable financial-domain abstractions and a pluggable calculation service layer, then demonstrate breadth through bounded prototype modules.
WS4	Risk Operations, Scenarios, Reconciliation & Reporting	Turn calculation services into a governed operational workflow with scenario comparison, traceability and evidence.
WS5	AI/GenAI, Automation & Innovation Layer	Add differentiated intelligence while keeping data access, financial calculations and consequential actions governed.

Recommended delivery order

• Foundation controls are cross-cutting and begin first: agree canonical data principles, data classification, permissions, audit, schema versioning and change control.

• Build the ingestion/mapping workbench against provisional synthetic source profiles because the client's real source shapes and attributes are not yet confirmed.

• Implement the configurable financial framework and one bounded liquidity vertical slice; show other financial modules as registered extension points until requirements and benchmark cases are available.

• Build run orchestration, scenarios, lineage and reconciliation as reusable services, not features hardcoded inside a single calculation.

• Add AI/GenAI with bounded tasks, evidence grounding, permissions, evaluation and human approval; external APIs are optional, not a hard dependency.

• Before production, complete client-specific data discovery, domain-expert validation, security testing, performance/resilience, migration and regulatory sign-off.

Shared synchronization contract

• Phase IDs P0–P6 and workstream IDs WS1–WS5 remain consistent across paired documents.

• Feature IDs are paired across FE and BE and used in tickets, tests and release notes.

• All proposed endpoint names must be confirmed in one OpenAPI specification before coding against them.

• Unknown source attributes are a requirements uncertainty, not a reason to hardcode a guessed client schema. Use versioned mappings, configurable extensions, profiling and review.

• Data security and governance are architecture-wide requirements, not an optional workstream that can be deferred.

• Every financial result is reproducible from pinned data and configuration versions; no AI-generated result is authoritative.